Stabilization baseline and release gates

SolariNet and RackWire | 20 October 2026 | operational snapshot, not a release declaration

The existing August 9 stabilization workflow remains the controlling plan. This snapshot records what was checked on October 20, the first integration changes, and the gates that still prevent a trustworthy platform release.

> **Release state:** No new SolariNet or RackWire version is ready to tag. SolariNet has no release tags; the candidate merges two unpublished stabilization commits with current main, updates the RackWire gitlink, and removes service-worker API caching. Security and durability gates remain open.

1. Observed state

Surface	Evidence from this session	Implication
Repositories	**OBSERVED** SolariNet GitHub and Forgejo main both resolve to `5e8d688`. Local stabilization has two commits ahead of its base and 31 mainline commits to reconcile. RackWire Forgejo main resolves to `5f29f59`; SolariNet still pins `fccf5ff` on published main.	Keep the integration in a review branch until its tests and cross-app behavior pass.
Live services	**OBSERVED** xenon server, client, monitor, dashboard, SoR sync, and RackWire discovery services are active. Dashboard and RackWire discovery health respond HTTP 200.	An August 8 maintenance handoff is historical; do not infer current health from it.
Dashboard static deploy	**OBSERVED** The served `api.jsx` alone differed from the repository and lacked the required CSRF header. The repository file was atomically installed on xenon, its rollback was rehearsed, and the served SHA-256 now matches `ce0861ef...`.	Authenticated writes may now work; an operator-session browser test remains required.
RackWire	**OBSERVED** Standalone current main passes 45 discovery tests. The first SolariNet integration run failed because its token test expected two retired `--sn-signal-*` colors. The candidate test now checks the contract's `--rw-cable-*` brand literals and passes.	The pin can advance only with cross-app smoke, including iframe, persistence, and live telemetry.

Surface	Evidence from this session	Implication
DNS and certificates	**OBSERVED** `rackwire.akoria.net` is NXDOMAIN locally and publicly, while the canonical `dashboard.akoria.net/rackwire/` path works. Public `akoria.net` certificate expired July 19; certbot fails ACME challenges.	Decide whether the separate RackWire hostname is a published surface. Repair public ACME reachability or use DNS-01 before calling the public site healthy.

2. Verification completed

- **OBSERVED** The merged candidate builds with server, I/O, SQLite, and JSON enabled; all 31 C test suites pass.
- **OBSERVED** PHP route/authorization tests pass; the full dashboard PHP and JS suites pass after the RackWire token-contract correction.
- **OBSERVED** A disposable MariaDB schema accepted every control-plane migration, and all 11 live DB integration cases passed, including request-queue and durable report-replay behavior. The schema was dropped afterward.
- **OBSERVED** A service-worker harness proved that offline API requests reject instead of reading an old authenticated cache, and activation removes the prior v6 API cache.
- **OBSERVED** The F-11 control bridge candidate now requires a complete newline-terminated request within a 16 KiB cap and two-second deadline. The PHP and C clients loop on short writes and reject incomplete replies. A socket-pair test exercises fragmented, 8 KiB, and truncated requests; C suites and PHP parser checks pass.
- **OBSERVED** The F-08 SoR candidate confirms mandatory broker routing before checkpointing, acknowledges consumer deliveries only after successful primary target apply, and reconciles periodically. Nineteen fault and ownership tests pass. F-08 remains open: live broker/crash drills and durable retry for the steel DNS copy are outstanding.
- **OBSERVED** The F-12 migration candidate installed 22 monitoring and 6 SoR files in disposable MariaDB 11.8 databases, then passed clean rechecks. It refused an untracked populated database and an interrupted ledger row. The former `--init-db` snapshot path now fails explicitly. Production adoption remains open; see the standalone migration runbook and PDF.
- **OBSERVED** The F-14 auth candidate gates local password login at the API and requires a private rate-limit directory; negative tests cover lockout, stale OIDC state, missing token claims, session expiry, and local role downgrade. The complete dashboard PHP suite passes. Live SSO callback, break-glass operation, and browser session tests remain open.
- **OBSERVED** The F-13 freshness candidate marks retained real data stale after a failed poll, shows its age, and pauses writes until recovery. Browser-adapter simulation verifies outage and recovery. The former SSE route immediately returns 204, releasing the PHP worker; the dashboard continues its 10-second poll. Authenticated browser observation remains open.
- **OBSERVED** F-10 candidates persist full client config before epoch activation and commit report receipts with current/history rows. Fault tests cover failed saves; live DB tests prove replay after reconnect/window eviction, sender sequence reuse, failed-write retry, and single monitor history writes. F-10 remains open for durable transport acknowledgments, coordinated fleet rollout, and Windows durability.
- **OBSERVED** Five DNS generation tests prove stable serials, isolated output, and rejection of corrupt history/CNAME cycles; generated zones and combined config pass BIND validation. Atomic deployment, rollback, and independent steel retry remain open.

3. Priority sequence and executable gates

1. **Contain and authenticate.** Complete F-02/F-03/F-13/F-14; test viewer, operator, origin, token, and offline states in a real browser. Cut over the PHP/host queue only after the uid split and a rehearsed rollback. A green route test alone is insufficient.
2. **Make state durable.** Close F-08 event delivery, F-10 convergence truth, F-12 migration ledger, and F-17 validated DNS apply. Inject broker loss and consumer crashes against disposable targets. No event may be acknowledged before its required target state is durable.
3. **Prove identity and lease behavior.** Bind certificates to node IDs and roles (F-01), move long scans off the lease loop (F-05), then exercise two-server lease and bind-failure cases (F-09).
4. **Integrate deliberately.** Advance RackWire's pin, preserve its standalone SQLite/discovery boundary, and test the SolariNet iframe/API/SoR commit boundary. Add shared maturity, provenance, freshness, and entity navigation as specified by I0-I4 in the August workflow.
5. **Publish only after evidence.** Produce self-contained HTML user/developer manuals and parallel Letter PDFs for each app; review release artifacts and run restore/rollback drills. Merge to Forgejo main, mirror GitHub, deploy staged artifacts, verify actual services, then tag matching releases.

> **Working contract:** `docs/SolariNet_Stabilization_Integration_Workflow_2026-08-09.md` defines F-01 through F-31 and I0-I4. RackWire's current `docs/CONTRACT-RW.md` supersedes the two retired signal-token assertions. The stale root README phase table and netdb YAML-authority text are corrected in this candidate, pending merge.

4. Unverified and blocked

- **UNVERIFIED** Authenticated browser writes, an offline-to-online browser transition, queue cutover on xenon, RackWire iframe actions against live SolariNet, SoR propagation through all render targets, migration adoption on current databases, and restore from backups.
- **UNVERIFIED** Current cross-build matrix, fleet-scale monitoring, three-monitor failover, public ACME challenge success, and complete manual/PDF parity across subprojects.
- **OBSERVED** Claude CLI cross-lab review could not run because its OAuth session had expired. Independent review is still an open release gate for the new security and integration work.

Prepared from repository state, local tests, read-only live checks, and one reversible dashboard static-file repair. Assertions marked Observed were exercised; the remaining release criteria require fresh evidence.